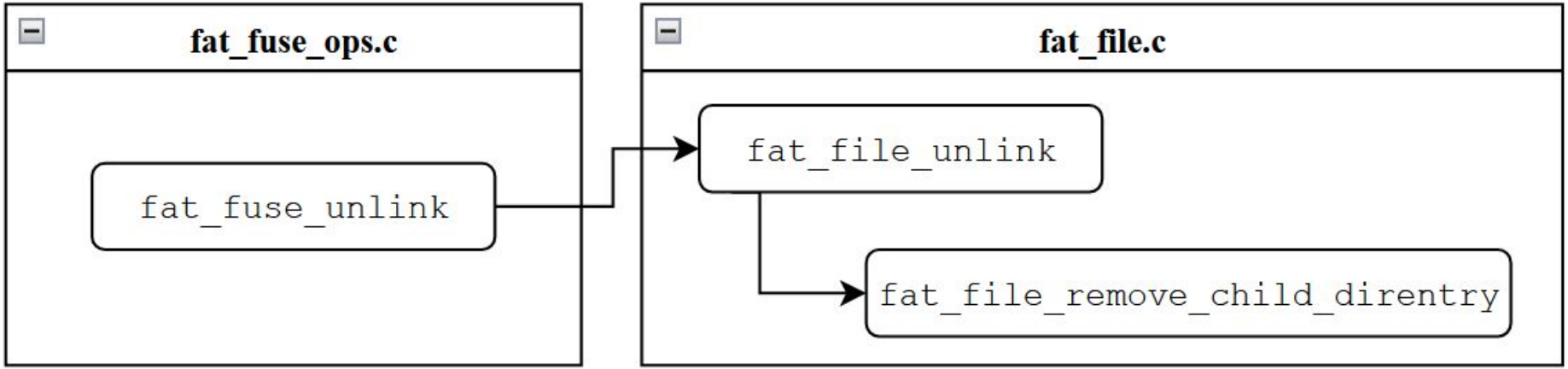
Borrado de archivos en FAT

SistOp - Lab 3

Borrado de archivos



Unlink de un file

```
def fat_file_unlink(file, parent):
    table = file.table
    cluster = file.start_cluster

    while not is_end_of_cluster(cluster):
        siguiente = table.next_cluster[cluster]
        table.next_cluster[cluster] = FAT_CLUSTER_FREE
        cluster = siguiente

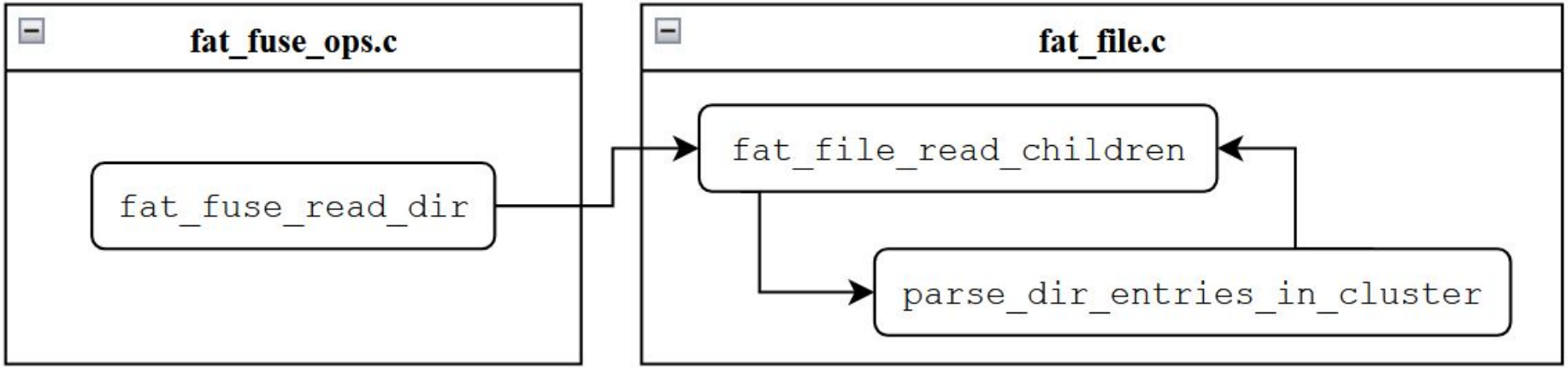
    fat_file_remove_child_direntry(parent, file.dentry)
```

Removiendo la direntry

```
def fat_file_remove_child_direntry(parent, child_dentry):
    child_dentry.base_name[0] = FILE_FOR_DELETION # 0xe5
    write_dir_entry(parent, child_dentry)
```

Cómo afecta esto en la lectura de un directorio?

Lectura de directorios



Los hijos de un directorio

```
def fat_file_read_children(dir):
    entry_list = []

    for cluster in dir.clusters():
        parse_dir_entries_in_cluster(cluster, entry_list)

    return entry_list
```

Parseo de dir entries

```
def parse_dir_entries_in_cluster(cluster, entry_list):
    for dentry in cluster.dir_entries():
        if is_end_of_directory(dentry):
            break
        if ignore_dentry(dentry):
            continue
        entry_list.append(dentry)
```

Predicados de dir entries

```
def is_end_of_directory(dentry):
    return dentry.base_name[0] == DENTRY_END_OF_DIR # 0x00

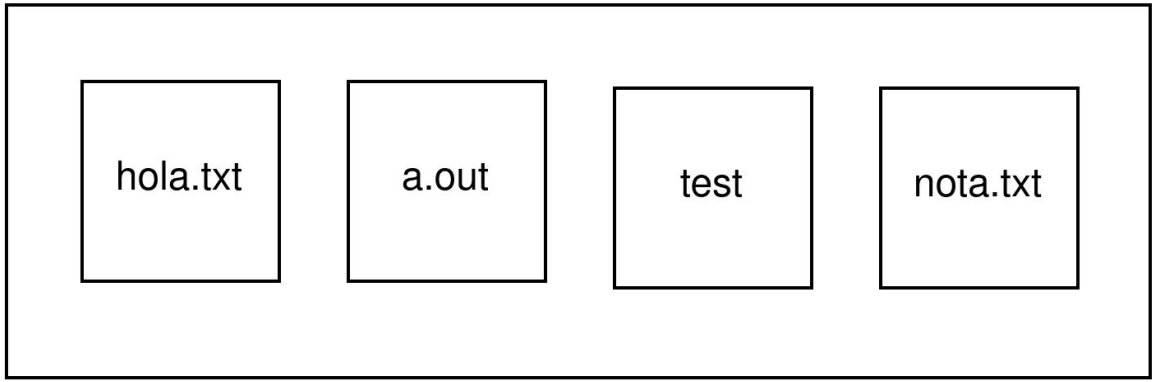
def ignore_dentry(dentry):
    return !file_basename_valid(dentry.base_name) ||
        !file_extension_valid(dentry.extension)

def file_basename_valid(base_name):
    if base_name[0] in {'\0', ' ', 0xe5}:
        return false

    ...
```

Parseo de dir entries

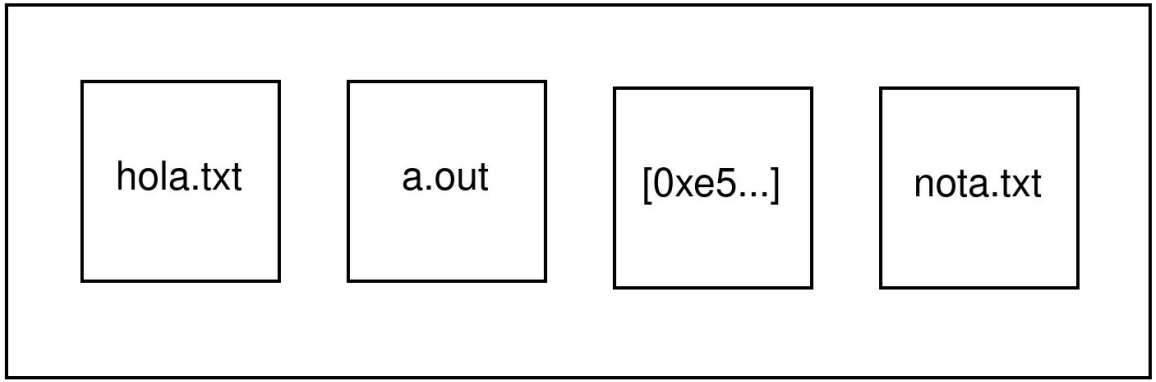
Caso normal: todas las dentries ocupadas por archivos “válidos”




```
entry_list = [hola.txt, a.out, test, nota.txt]
```

Parseo de dir entries

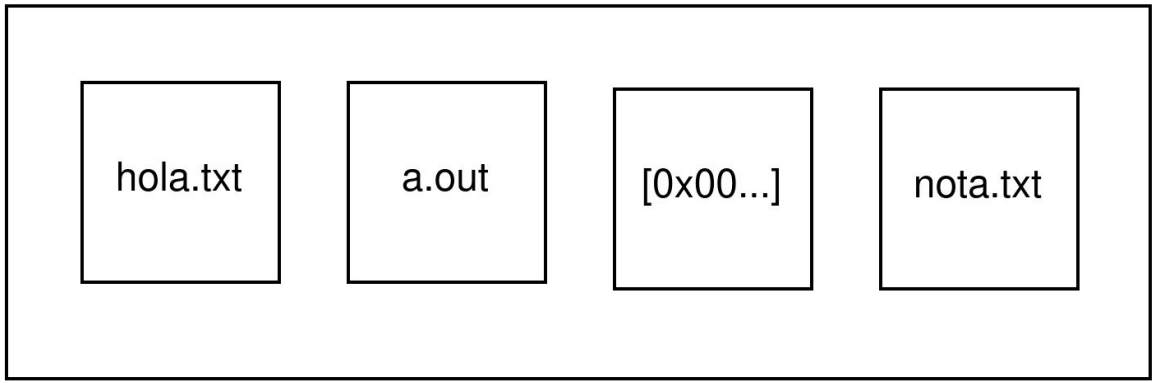
Caso normal: una dentry marcada como `FILE_FOR_DELETE`




```
entry_list = [hola.txt, a.out, nota.txt]
```

Parseo de dir entries

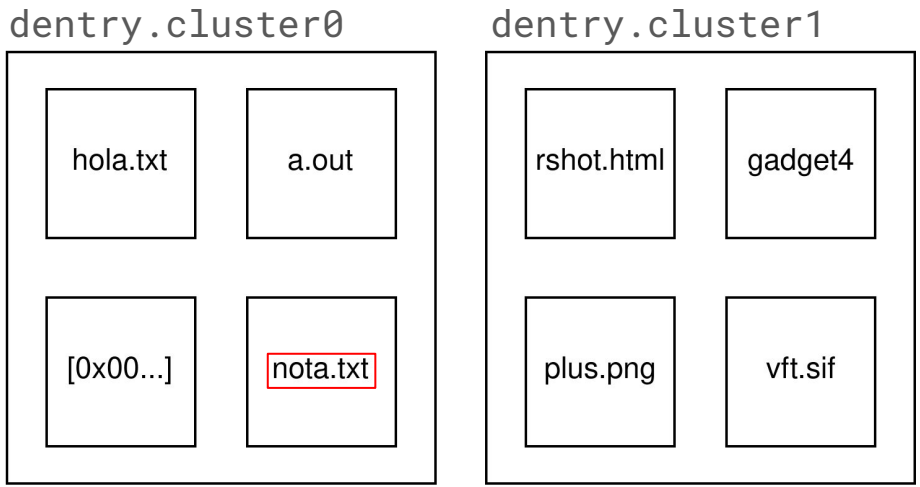
Caso erroneo: una dentry marcada con ceros




```
entry_list = [hola.txt, a.out]
```

Parseo de dir entries

Caso borde: múltiples clusters y una dentry marcada con ceros




```
entry_list = [hola.txt, a.out, rshot.html,
    gadget4, plus.png, vft.sif]
```